

Task 4 – Maintaining, Supporting, and Testing a Procedural Programming Application

1. Introduction

The payroll processing application was tested manually by running the program many times and entering different inputs across different scenarios and observing the results in the terminal.

The aim of this testing was to ensure the system handles undesired user inputs and guides the user where necessary, ensuring that each screen produces what is required in the design stage.

2. Test Table

Testing payslip calculations

Test ID	Feature	Input Type	Input	Expected Output	Actual Output	Pass/Fail
T01	Monthly gross	Normal	54000	4500.00	4500.00	Pass
T02	Monthly PAYE	Boundary	44000	420.83	420.83	Pass
T03	Monthly USC	Boundary	13000	0.00	0.00	Pass
T04	Monthly PRSI	Boundary	18304	0.00	0.00	Pass

Testing input validations

Test ID	Feature	Input Type	Input	Expected Output	Actual Output	Pass/Fail
T05	Employee ID format	Invalid	ab1	Invalid input message: Not a valid ID. Try again or press (c) to cancel.	Same as expected	Pass
T06	Employee ID exists	Invalid	id999	ID error message: This ID does not exist. Try again or press (c) to cancel.	Same as expected	Pass
T07	Menu option	Invalid	5	Invalid input message: Invalid input. Press 1, 2, 3 or 4...	Same as expected	Pass
T08	Press enter to go back	Normal	Enter	Returns to profile	Program ended abruptly	Fail

Test ID	Feature	Input Type	Input	Expected Output	Actual Output	Pass/Fail
T09	Name too long	Invalid	Bartholomew Vanderburg	Input too long error: Name is too long. Enter 20 characters or less:	Same as expected	Pass
T10	Salary data	Invalid	asd	Invalid input error: Invalid input. Press (c) to cancel or enter a valid salary amount. Salary can only be numeric and 5 or 6 digits	Validation works correctly but the error message has copy paste error: Role is too long. Enter 20 characters or less	Fail
T11	Salary data	Boundary	10000	Salary detail is updated	Same as expected	Pass
T12	PPSN format	Invalid	123456a	Invalid input message: Enter a valid PPSN:	Same as expected	Pass
T13	PPSN exists	Invalid	1234567a	PPSN error message: This PPS number is already in the system.	Same as expected	Pass

Testing add/remove/edit employee

Test ID	Feature	Input Type	Input	Expected Output	Actual Output	Pass/Fail
T14	Add employee	Normal	jane doe, it, developer, 35000, 9876543z	Record added	Same as expected	Pass
T15	Add another employee	Normal	john doe, it, developer, 35000, 8976543z	Record added	Same as expected	Pass
T16	Change mind before removing employee	Normal	Select n	Returns to profile	Same as expected	Pass

Test ID	Feature	Input Type	Input	Expected Output	Actual Output	Pass/Fail
T17	Remove employee	Normal	Select y	Record deleted	Same as expected	Pass
T18	Change mind before editing salary	Normal	Select c	Profile remains unchanged	Same as expected	Pass
T19	Edit salary	Normal	Enter 54090	Salary updated	Same as expected	Pass

Testing display payslip/profile

Test ID	Feature	Input Type	Input	Expected Output	Actual Output	Pass/Fail
T20	Display payslip	Normal	Select display payslip	Payslip displays in correct format and correct data	Same as expected	Pass
T21	Display profile	Normal	Enter ID1	Alice Murphy's profile shows with full details	Same as expected	Pass

3. Recommendations

These improvements are suggested for the program:

- The Help menu needs to be expanded. Currently, there is a placeholder for the Help content. The content can be organized under titles for each menu, explaining what each option does.
- The requirements state that this application should support up to 50 employees but there is no enforcement in the source code to limit the number of employee records stored in employees.json.
- There is an input validation function checking the salary input format. However, there is no validation that checks realistic amounts or minimum wage amount. As a result, a user can add a very low annual salary (e.g. €10000) for an employee and the system will accept it without any warning.
- There is an input validation function checking the user inputs in the Add New Employee screen. Currently, these validations enforce a limit of 20 characters. This may not be realistic for some names where the last name is really long or when the employee has more than one name (e.g. a middle name).
- It would be a better idea to select department and role from a pre-set list of departments and roles when adding a new employee. This would avoid inconsistent data input such as "IT" and "Information Technologies" which actually point to the same department.
- Currently, there is no way to go back to a previous menu in the Add New Employee screen without adding the details in the correct format. If a user changes their mind and wants to cancel filling in the new employee form, they have no option but to enter dummy info, save it in the system and then go back to remove the same record from the system.

4. Supporting Documentation

User Guide

- To run the program:
 1. Ensure Python 3.12 or higher is installed on the machine.
 2. Ensure all of the source files, including the employees.json, are in the same directory.
 3. Click the Play button at the top of the file in Visual Studio Code. Alternatively, you can start the program from the terminal by typing `python3 payroll_program.py` in the directory where the source file is.
- Menu options accept appropriate keyboard inputs followed by pressing Enter.
- Viewing the output in a large monitor is recommended for a better experience. Smaller screens introduce the risk of terminal output being wrapped.

Code Documentation

- **payroll_program.py:** This is where the program starts. It imports payslip_calculations and input_validations. All screen functions are contained in this file.
- **payslip_calculations.py:** This file includes all the functions to make salary and tax calculations.
- **input_validations.py:** User input validations are handled by the functions in this file.

Assumptions/Requirements

- This application should be run with a Python version of 3.12 or higher. Older versions do not support single quotes in f-strings which this program relies on to run smoothly.
- The formatting of the payslip and the employee profiles were designed with small laptop screens in mind. However, if the terminal screen is too small, the formatting might not look as designed, some lines may be wrapped or truncated.